SAJAL MALHOTRA

+91-62845-96119 | malhotrasajal243@gmail.com | linkedin.com/in/sajal-malhotra1 | github.com/SAJAL-MALHOTRA

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology Jalandhar **Aug 2025 – Present**

B.Tech in Information Technology — CGPA: 8.13 *Jalandhar, Punjab*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Machine Learning, Computer Networks

TECHNICAL PROJECTS

ThinkFolio — Adaptive ML Reasoning Engine | *PyTorch, XGBoost, FastAPI, PostgreSQL, Docker, Next.js*

- Engineered a dual-branch ML evaluation engine (PyTorch Tabular MLP + gradient-boosted trees) scoring financial reasoning across 8 dimensions, reaching **R^2 = 0.927** predictive accuracy.
- Integrated **FinBERT** sentence embeddings with SHAP explainability to deliver real-time feature attribution at under **100ms** latency overhead.
- Architected a containerized microservice backend with **FastAPI**, **SQLAlchemy**, and **PostgreSQL**, adding EMA-based competency tracking and Alembic migrations.
- Built an interactive evaluation UI in **Next.js** and **Tailwind CSS** that surfaces dynamic rubric scores and personalized learning trajectories.

Pairfect — Real-Time Interactive Consumer Platform | *Next.js, Supabase, Groq API (Llama 3), Tailwind CSS*

- Designed and shipped a full-stack real-time social app (**pairfect.site**) with live state synchronization and LLM-driven prompt analysis.
- Built a **Groq-accelerated Llama 3** inference pipeline delivering sub-**50ms** evaluation latency for instant conversational scoring and "vibe judging."
- Implemented real-time bidirectional communication using **Supabase WebSockets** with PostgreSQL Row-Level Security (RLS) policies.
- Enforced end-to-end **TypeScript** schemas across all real-time payloads, shipping with zero TypeScript errors across 77 routes.

ScholarAI / ResearchFlow — Stateful Multi-Agent Engine | *LangGraph, LangChain, Python, FastAPI, pgvector*

- Architected a **5-node stateful LangGraph pipeline** (Planner → Researcher → Evidence Collector → Verifier → Writer) automating academic paper discovery, semantic parsing, and structured synthesis.
- Implemented vector similarity search over **Supabase pgvector** with semantic embeddings for high-precision retrieval across technical literature.
- Built asynchronous REST APIs with **FastAPI** to handle long-running multi-agent reasoning cycles with checkpointing and state recovery.

Computer Vision Illumination Classifier | *PyTorch, Torchvision, OpenCV, Python*

- Built a deep CNN ensemble (**ResNet-18/34**, **EfficientNet-B0**, handcrafted illumination features, FastMLP) for 3-class surveillance illumination classification, achieving **94.2%** validation accuracy.
- Engineered an end-to-end pipeline with custom data loaders, geometric augmentations, and evaluation metrics; ranked in the **top 7% (169/227)** of a national computer vision challenge.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, Bash

ML & AI: PyTorch, XGBoost, scikit-learn, LangGraph, LangChain, HuggingFace (FinBERT), OpenCV

Backend & Web: FastAPI, Next.js, React, Node.js, Express.js, Tailwind CSS, REST APIs

Databases & Storage: PostgreSQL, Supabase, pgvector, Redis, SQLite, SQLAlchemy

Tools & Infrastructure: Docker, Git/GitHub, Linux (Arch Linux), Postman, Alembic, CI/CD

ACHIEVEMENTS & EXTRACURRICULAR

- **Top 7% (169/227)** in a national computer vision challenge on illumination prediction from camera images.
- **Core Member, Cybernauts, NIT Jalandhar** — organized coding competitions and technical events, engaging students in competitive programming and software development.
- **Content Creator & Model** — built a personal Instagram presence generating **2.2L+ (220K+)** monthly views across fashion, tech, and creative content.
- **Fashion & Modelling Club, NIT Jalandhar** — participated in modelling events, workshops, shoots, and stage performances.